Semester I Examinations 2015/2016

Exam Codes	3BA1, 3BS9, 3MR3, 3BPT1,4BME1 1OA1, 1EM1
Exams	Third Arts, Third Science, Third Marine Science, Fourth Mathematics and Education,
Module	Applied Statistics I
Module Codes	**ST311** **MA5103**
External Examiner	Professor S. T. Buckland
Internal Examiner(s)	Dr. E. Holian Professor C. Seoighe
Instructions:	**Answer any 3 questions from Questions 1-4** **All Questions carry equal marks.**
Duration	2 Hours
No. of Pages	19 pages, including this one
School	School of Mathematics, Statistics and Applied Mathematics
Release to Library:	No

Requirements:

Statistical Tables/ Log Tables	Relevant distribution tables are attached to this paper. The *New Formulae & Tables* are optional.
Other Materials Allowed	A calculator is allowed (non-programmable and not capable of storing text).

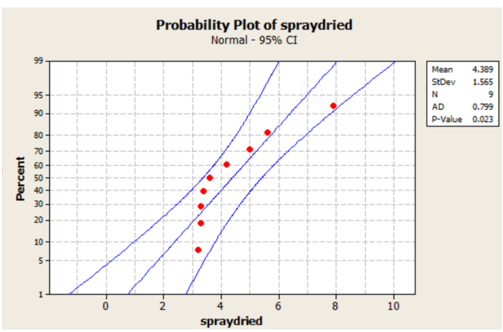
Question 1

Complete parts (a), (b) and (c).

(a) Instant coffee can be formulated via different methods - freeze-drying, spray-drying etc. From a health standpoint, the amount of caffeine that is left as residue is an important factor. A sample of spray-dried coffee brands were observed to have the following residue in grams (per 100 grams of dry weight):
3.2, 3.3, 3.3, 3.4, 3.6, 4.2, 5.0, 5.6, 7.9.
A summary of descriptive statistics and plots for this sample are provided below:

Variable	N	N*	Mean	SE Mean	StDev	Minimum	Q1	Median	Q3	Maximum
spraydried	9	0	4.389	0.522	1.57	3.20	3.30	3.60	5.30	7.90

Variable	Mode	N for Mode	Skewness	Kurtosis
spraydried	3.3	2	1.66	2.60



(i) By comparing the values of the mean, the median and the mode(s) for this sample what does this tell us about the shape of the distribution of the sample? [2 marks]

(ii) What does the normal probability plot suggest? [2 marks]

(iii) Construct the letter value display for this sample, including midvalues and spreads. [4 marks]

(iv) What do the letter values indicate about the sample distribution? [4 marks]

(b) The researcher wishes to determine whether this sample gives evidence that the typical amount of residue for the population of all spray-dried coffees is greater than 3.4 grams.

(i) It is suggested that a 1-sample t-test be carried out. The output is included below. Assuming the assumptions of the test hold what can you conclude as a result of this analysis? Reference the output to justify your answer. [3 marks]

Test of mu = 3.4 vs > 3.4

Variable	N	Mean	StDev	SE Mean	95% Lower Bound	T	P
spraydried	9	4.389	1.565	0.522	3.419	1.90	0.047

Question 1 continues on the next page

(ii) Do you have any concerns about applying the 1-sample t-test in this application? If so, explain reasons for your concern. [2 marks]

(iii) It is suggested that a 1-sample Sign test be carried out. The output is included below. Verify, by calculation, the value of the p-value as printed in the output for the Sign test. Include explanation of how the p-value is calculated defining any symbols you use. [4 marks]

```
Sign test of median =  3.4 versus > 3.4

              N  Below  Equal  Above       P  Median
spraydried    9      3      1      5   0.363    3.60
```

(iv) If testing at $\alpha = 0.05$, what can you conclude as a result of the Sign test? Include a statement of the hypotheses being tested, your decision and conclusion in this application. [3 marks]

(c) The researcher has developed a new method of drying instant coffee and takes a sample of coffees dried via this new method again observing the residue in grams (per 100 grams of dry weight). The observed amounts of residue for the spray-dried sample and the sample of coffees dried via the new method are provided below with some summary statistics.

spraydried	3.2, 3.3, 3.3, 3.4, 3.6, 4.2, 5.0, 5.6, 7.9.
new method	3.0, 3.7, 5.1, 5.6, 7.2, 8.0, 8.1, 8.4.

```
Descriptive Statistics:

Variable     N  N*   Mean  StDev  Median
spraydried   9   0  4.389  1.565    3.60
new method   8   0  6.137  2.095    6.40
```

The researcher wishes to compare the typical amount of residue for the population of all spray-dried coffees to the typical amount of residue for the population of coffees dried via the new method, in particular whether the residues for spray-dried coffees are less than residues for coffees dried via the new method.

(i) It is suggested to you to carry out a Wilcoxon Rank Sum test. Explain why, in this application, this test is more suitable than the 2-sample Z test, or the 2-sample t test. [4 marks]

(ii) Carry out the test at significance level α=0.05. Include a statement of the hypotheses being tested, formulation of the test statistic, the rejection region for the test statistic, and indicate clearly your decision and conclusion from the result of the test. [7 marks]

[Total 35 marks]

Question 2

Complete parts (a), (b) and (c).

(a) (i) *"The true correlation coefficient between two variables X and Y for a population of individuals is known to have value $\rho = 0$. This implies that X and Y are unrelated."*
Is this statement true or false? If you deem the statement to be false give a reason for your answer. [2 marks]

(ii) A sample of 25 individuals gave a sample correlation coefficient of $r = 0.64$. Carry out a hypothesis test of whether there is evidence of a significant increasing relationship between X and Y in the population. Clearly state the null and alternative hypotheses of your test, and give a test statistic, a rejection region, your decision and your conclusion. [6 marks]

(b) Consider a simple linear regression model $Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$.

(i) What does the error term, ϵ_i, represent? [2 marks]

(ii) When carrying out an Analysis of Variance, explain in general what the terms SS_{Total}, SS_{Reg} and SS_{Error} represent. [5 marks]

(c) The following model is being considered in order to model the variability in a response variable Y, using five candidate predictor variables, X_1, X_2, X_3, X_4, X_5,

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_3 X_{3i} + \beta_4 X_{4i} + \beta_5 X_{5i} + \epsilon_i.$$

The model is fitted to observations obtained for a sample of 25 individuals, and the results of the fit are provided in the partial printout on the next page.
Use the output to answer the following questions.

(i) Calculate the coefficient of determination R^2 for this regression fit. Interpret the meaning of this value. [2 marks]

(ii) Find the missing values in the ANOVA table, and complete the test, at the $\alpha = 0.05$ level, including statements of the hypotheses being tested, a rejection region, a decision and conclusion. (*You do not have to calculate the p-value.*) [6 marks]

(iii) Complete the missing t-test in the output, i.e. test whether X_3 is a good predictor of Y at the $\alpha = 0.05$ level. Include the hypotheses being tested, the calculation of the test statistic, a rejection region, a decision and conclusion. [5 marks]

(iv) Use the partial F-test to test the hypothesis : $H_0 : \beta_4 = \beta_5 = 0$ at $\alpha = 0.05$. Interpret your findings. [5 marks]

(v) The output provides a prediction interval and a confidence interval for the response when $X_1 = 25$, $X_2 = 13$, $X_3 = 1$, $X_4 = 66$, $X_5 = 9$. Explain the difference as to when a prediction interval and a confidence interval applies. [2 marks]

Question 2 continues on the next page

Question 2 continued...

```
The regression equation is
Y = 33.0 + 7.44 X1 - 6.73 X2 + 30.4 X3 + 0.196 X4 + 5.78 X5

Predictor     Coef  SE Coef      T      P
Constant     33.00    78.63   0.42  0.680
X1            7.44    2.987   2.49  0.022
X2           -6.73    0.831  -8.09  0.000
X3           30.39    22.15     -      -
X4           0.196    0.507   0.39     -
X5           5.78     1.785   3.24     -

S = 48.8882   R-Sq = - %   R-Sq(adj) = - %

Analysis of Variance

Source          DF      SS     MS      F      P
Regression       -  244404     -       -      -
Residual Error   -       -     -
Total            -  289815

Source  DF  Seq SS
X1       1   56378
X2       1  153278
X3       1    4599
X4       1    5064
X5       1   25084

Predicted Values for New Observations

NewObs     Fit  SE Fit       95% CI            95% PI
1       226.83   16.34  (192.64, 261.03)  (118.95, 334.72)

Values of Predictors for New Observations

NewObs    X1    X2    X3    X4    X5
1        25.0  13.0  1.00  66.0  9.00
```

[Total 35 marks]

Question 3

Complete parts (a), (b) and (c).

The manager at a stationery outlet is interested in estimating stocking levels of greeting cards. Data for a random sample of 28 greeting card designs were collected including the quantity of each design sold, *qty*, and four possible explanatory variables:

- *size*, size of card in cm^2, front cover area.
- *price*, price of card in euro.
- *occasion*, value 1 if the card is for celebrating a birthday, or value 0 for any other occasion.
- *gender*, value 1 if the card is gender neutral, or value 0 if gender specific.

(a) A multiple linear regression model using three of the explanatory variables, *size*, *occasion* and *gender*, was first considered giving the following output:

```
The regression equation is
qty = 56.01 + 0.29 size + 6.00 occasion - 1.04 gender

Predictor     Coef  SE Coef      T      P
Constant     56.01    7.345   7.63  0.000
size          0.29   0.0310   9.36  0.000
occasion      6.00    2.496   2.40  0.024
gender       -1.04    2.364  -0.44  0.664

S = 5.57578   R-Sq = 79.1%   R-Sq(adj) = 76.5%

Analysis of Variance

Source          DF       SS      MS      F      P
Regression       3  2827.46  942.49  30.32  0.000
Residual Error  24   746.14   31.09
Total           27  3573.60

Source    DF   Seq SS
size       1  2642.66
occasion   1   178.78
gender     1     6.02
```

(i) Interpret the meaning of the estimated coefficients of the three predictor variables *size*, *occasion* and *gender*. [5 marks]

(ii) Write down the hypotheses, in terms of the model coefficients, β's, being tested in the ANOVA table for this model.
What do you conclude as a result of this test:
- in terms of the model coefficients, β's, and
- in terms of the predictor variables? [5 marks]

(iii) Comment on the significance of the three predictor variables *size*, *occasion* and *gender*. Use the output to support your answer. Include a statement of the hypotheses being tested for any hypothesis test you reference. [6 marks]

Question 3 continues on the next page

Question 3 continued...

(b) Three models involving the predictors *size* and *price* are fitted to the data giving the following outputs:

Model 1:

```
The regression equation is
qty = 63.2 + 0.273 size

Predictor     Coef  SE Coef     T      P
Constant    63.186    6.756  9.35  0.000
size       0.27259  0.03173  8.59  0.000

S = 5.98376   R-Sq = 73.9%   R-Sq(adj) = 72.9%
```

Model 2:

```
The regression equation is
qty = 104 + 5.38 price

Predictor     Coef  SE Coef      T      P
Constant   104.129    2.145  48.55  0.000
price       5.3755   0.6070   8.86  0.000

S = 5.84977   R-Sq = 75.1%   R-Sq(adj) = 74.1%
```

Model 3:

```
The regression equation is
qty = 113 - 0.057 size + 6.48 price

Predictor     Coef  SE Coef      T      P
Constant    112.68    45.77   2.46  0.021
size       -0.0566   0.3028  -0.19  0.853
price        6.477    5.925   1.09  0.285

S = 5.96145   R-Sq = 75.1%   R-Sq(adj) = 73.1%

Analysis of Variance

Source          DF      SS      MS      F      P
Regression       2  2685.1  1342.6  37.78  0.000
Residual Error  25   888.5    35.5
Total           27  3573.6

Source  DF  Seq SS
size     1  2642.7
price    1    42.5
```

Discuss the predictive power of the three variables, by giving your conclusions from the outputs of the t-tests and ANOVA test provided. Ensure you provide the hypotheses being tested for any test that you reference. [8 marks]

Question 3 continues on the next page

(c) The following output was produced using the best subsets routine in which all four candidate predictors are considered.

(i) Explain the meaning of the phrase "a parsimonious model". [2 marks]

(ii) Write down the most suitable model, i.e. the set of predictors, that you think best explains the variability in observed `qty`.
Justify your choice of model by explaining the role of s, R^2, adjusted R^2, and Mallow's C_p in choosing the best model. [9 marks]

```
Response is qty

                                                o
                                                c
                                                c g
                                              p a e
                                            s r s n
                                            i i i d
                        Mallows             z c o e
Vars  R-Sq   R-Sq(adj)       Cp      S      e e n r
1     75.1        74.1      6.3  5.8498       X
1     73.9        72.9      7.7  5.9838     X
2     80.7        79.2      1.5  5.2497       X X
2     79.0        77.3      3.6  5.4851     X   X
3     80.9        78.5      3.2  5.3320     X X X
3     80.9        78.5      3.2  5.3327       X X X
4     81.1        77.8      5.0  5.4186     X X X X
```

[Total 35 marks]

Question 4

Complete parts (a), (b) and (c).

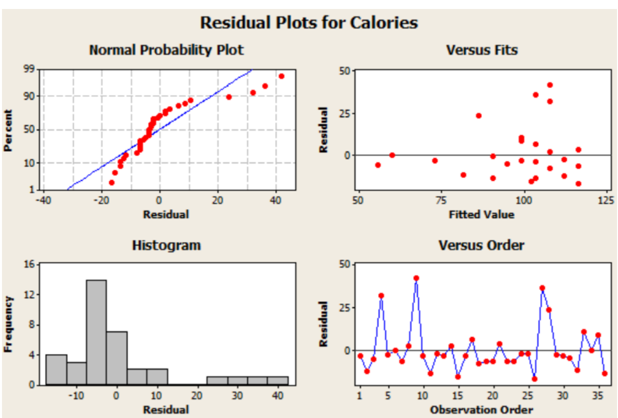
(a) When fitting a simple linear regression model to data what assumptions are required? [5 marks]

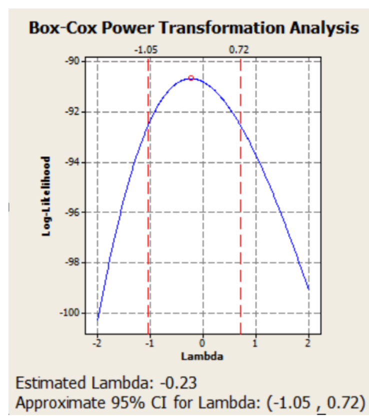
(b) The output and diagnostic plots provided below show the results of a simple linear regression fit to the number of calories (per serving), *Calories*, observed for a sample of 36 cereals, using the amount of fibre (grams per serving), *Fibre*, as a predictor.

```
The regression equation is
Calories = 117 - 4.36 Fibre

Predictor     Coef   SE Coef      T      P
Constant       117     3.288  35.50  0.000
Fiber        -4.36    0.6501  -6.71  0.000

S = 13.8903   R-Sq = 57.0%   R-Sq(adj) = 55.7%
```





(i) Comment on the validity of the assumptions for this regression fit. [6 marks]

(ii) What does the output of the Box-Cox transformation indicate? [5 marks]

(iii) A simple linear regression using *Fibre* as the predictor was fitted to the transformed response variable $ln(Calories)$ giving the output below:

```
The regression equation is
lnCalories = 4.78 - 0.052 Fibre

Predictor       Coef    SE Coef       T      P
Constant        4.78    0.03073  155.48  0.000
Fibre         -0.052   0.006076   -8.50  0.000

S = 0.129809   R-Sq = 68.0%   R-Sq(adj) = 67.0%
```

Interpret the meaning of the estimated coefficient of *Fibre* in this model. [5 marks]

Question 4 continues on the next page

Question 4 continued...

(iv) Using the model in part (iii) predict the number of calories of a cereal which has 4 grams per serving of fibre. [4 marks]

(c) The model fitted in (b) (iii), to *lnCalories*, also produced the following output:

```
Unusual Observations

Obs  Fibre  lnCalories     Fit  SE Fit  Residual  St Resid
 3   14.0      3.9120  4.0552  0.0668   -0.1432     -1.29 X
 4    2.0      4.9416  4.6746  0.0237    0.2671      2.09R
 9    2.0      5.0106  4.6746  0.0237    0.3360      2.63R
27    3.0      4.9416  4.6230  0.0219    0.3187      2.49R
28    7.0      4.7005  4.4165  0.0299    0.2840      2.25R
34   13.0      4.0943  4.1068  0.0611   -0.0125     -0.11 X

R denotes an observation with a large standardized residual.
X denotes an observation whose X value gives it large leverage.
```

(i) Explain why the observations marked with R are reported in the output above. [5 marks]

(ii) Explain why the observations marked with X are reported in the output above. [5 marks]

[Total 35 marks]

ST311 Formulae

Some useful z-scores:

$Z_{0.1} =$	1.28
$Z_{0.05} =$	1.64
$Z_{0.025} =$	1.96
$Z_{0.01} =$	2.33
$Z_{0.005} =$	2.58

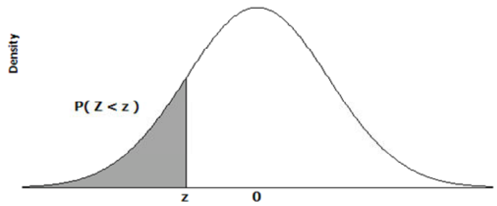
Some ranges for the standard normal distribution:

1.35	middle 50% : $Z_{0.75} - Z_{0.25}$
2.30	middle 75% : $Z_{0.875} - Z_{0.125}$
3.07	middle 87.5% : $Z_{0.9375} - Z_{0.0625}$
3.73	middle 93.75% : $Z_{0.96875} - Z_{0.03125}$
4.31	middle 96.875% : $Z_{0.984375} - Z_{0.015625}$
4.84	middle 98.4375% : $Z_{0.9921875} - Z_{0.0078125}$

Some formulae that may be useful when carrying out inference with parametric procedures:

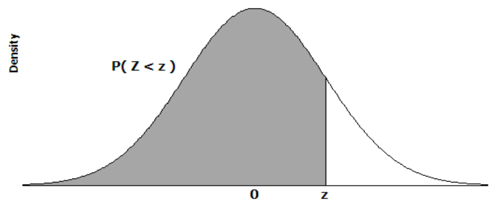
- Inference for population correlation coefficient ρ: $TS = r\sqrt{\frac{n-2}{1-r^2}}$
- Partial F-test : $TS = \frac{(\text{additional SSreg})/q}{MS_{error}}$

Table: Cumulative Probabilities for the Standard Normal Distribution, Z~Normal (0,1).



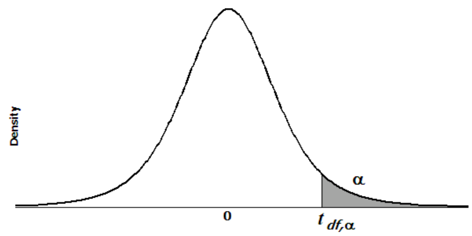
Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
-3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
-3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
-3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
-3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
-3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
-2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
-2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
-2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
-2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
-2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
-2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
-2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
-2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
-2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
-2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
-1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
-1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
-1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
-1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
-1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
-1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
-1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
-1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
-1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
-1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
-0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
-0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
-0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
-0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
-0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
-0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
-0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
-0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
-0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
-0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641

Table: Cumulative Probabilities for the Standard Normal Distribution, Z~Normal (0,1).



Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998

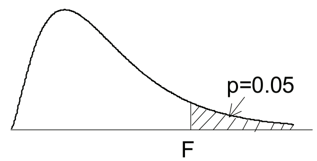
Table: t distribution critical values



df	Upper tail probability 0.25	0.20	0.15	0.10	0.05	0.025	0.02	0.01	0.005	0.0025	0.001	0.0005
1	1.000	1.376	1.963	3.078	6.314	12.706	15.895	31.821	63.657	127.321	318.309	636.619
2	0.816	1.061	1.386	1.886	2.920	4.303	4.849	6.965	9.925	14.089	22.327	31.599
3	0.765	0.978	1.250	1.638	2.353	3.182	3.482	4.541	5.841	7.453	10.215	12.924
4	0.741	0.941	1.190	1.533	2.132	2.776	2.999	3.747	4.604	5.598	7.173	8.610
5	0.727	0.920	1.156	1.476	2.015	2.571	2.757	3.365	4.032	4.773	5.893	6.869
6	0.718	0.906	1.134	1.440	1.943	2.447	2.612	3.143	3.707	4.317	5.208	5.959
7	0.711	0.896	1.119	1.415	1.895	2.365	2.517	2.998	3.499	4.029	4.785	5.408
8	0.706	0.889	1.108	1.397	1.860	2.306	2.449	2.896	3.355	3.833	4.501	5.041
9	0.703	0.883	1.100	1.383	1.833	2.262	2.398	2.821	3.250	3.690	4.297	4.781
10	0.700	0.879	1.093	1.372	1.812	2.228	2.359	2.764	3.169	3.581	4.144	4.587
11	0.697	0.876	1.088	1.363	1.796	2.201	2.328	2.718	3.106	3.497	4.025	4.437
12	0.695	0.873	1.083	1.356	1.782	2.179	2.303	2.681	3.055	3.428	3.930	4.318
13	0.694	0.870	1.079	1.350	1.771	2.160	2.282	2.650	3.012	3.372	3.852	4.221
14	0.692	0.868	1.076	1.345	1.761	2.145	2.264	2.624	2.977	3.326	3.787	4.140
15	0.691	0.866	1.074	1.341	1.753	2.131	2.249	2.602	2.947	3.286	3.733	4.073
16	0.690	0.865	1.071	1.337	1.746	2.120	2.235	2.583	2.921	3.252	3.686	4.015
17	0.689	0.863	1.069	1.333	1.740	2.110	2.224	2.567	2.898	3.222	3.646	3.965
18	0.688	0.862	1.067	1.330	1.734	2.101	2.214	2.552	2.878	3.197	3.610	3.922
19	0.688	0.861	1.066	1.328	1.729	2.093	2.205	2.539	2.861	3.174	3.579	3.883
20	0.687	0.860	1.064	1.325	1.725	2.086	2.197	2.528	2.845	3.153	3.552	3.850
21	0.686	0.859	1.063	1.323	1.721	2.080	2.189	2.518	2.831	3.135	3.527	3.819
22	0.686	0.858	1.061	1.321	1.717	2.074	2.183	2.508	2.819	3.119	3.505	3.792
23	0.685	0.858	1.060	1.319	1.714	2.069	2.177	2.500	2.807	3.104	3.485	3.768
24	0.685	0.857	1.059	1.318	1.711	2.064	2.172	2.492	2.797	3.091	3.467	3.745
25	0.684	0.856	1.058	1.316	1.708	2.060	2.167	2.485	2.787	3.078	3.450	3.725
26	0.684	0.856	1.058	1.315	1.706	2.056	2.162	2.479	2.779	3.067	3.435	3.707
27	0.684	0.855	1.057	1.314	1.703	2.052	2.158	2.473	2.771	3.057	3.421	3.690
28	0.683	0.855	1.056	1.313	1.701	2.048	2.154	2.467	2.763	3.047	3.408	3.674
29	0.683	0.854	1.055	1.311	1.699	2.045	2.150	2.462	2.756	3.038	3.396	3.659
30	0.683	0.854	1.055	1.310	1.697	2.042	2.147	2.457	2.750	3.030	3.385	3.646
40	0.681	0.851	1.050	1.303	1.684	2.021	2.123	2.423	2.704	2.971	3.307	3.551
50	0.679	0.849	1.047	1.299	1.676	2.009	2.109	2.403	2.678	2.937	3.261	3.496
60	0.679	0.848	1.045	1.296	1.671	2.000	2.099	2.390	2.660	2.915	3.232	3.460
80	0.678	0.846	1.043	1.292	1.664	1.990	2.088	2.374	2.639	2.887	3.195	3.416
100	0.677	0.845	1.042	1.290	1.660	1.984	2.081	2.364	2.626	2.871	3.174	3.390
1000	0.675	0.842	1.037	1.282	1.646	1.962	2.056	2.330	2.581	2.813	3.098	3.300
Z*	0.674	0.842	1.036	1.282	1.645	1.960	2.054	2.326	2.576	2.807	3.090	3.291

F distribution for p = 0.05

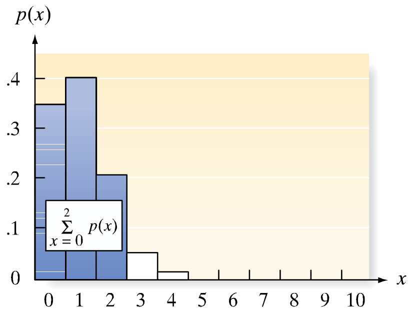
For various pairs of degress of freedom (v_1,v_2), the tabled entries represent the critical values of F above which the proportion p = 0.05 of the distribution falls. (Example, for df = 3,8 and F = 2.92 is exceeded by p = 0.05 of the population.



***p* = 0.05**

	Degrees of freedom for the numerator v_1																	
v_2	1	2	3	4	5	6	7	8	9	10	12	15	20	30	40	60	120	∞
1	161.446	199.499	215.707	224.583	230.160	233.988	236.767	238.884	240.543	241.882	243.905	245.949	248.016	250.096	251.144	252.196	253.254	254.317
2	18.513	19.000	19.164	19.247	19.296	19.329	19.353	19.371	19.385	19.396	19.412	19.429	19.446	19.463	19.471	19.479	19.487	19.496
3	10.128	9.552	9.277	9.117	9.013	8.941	8.887	8.845	8.812	8.785	8.745	8.703	8.660	8.617	8.594	8.572	8.549	8.526
4	7.709	6.944	6.591	6.388	6.256	6.163	6.094	6.041	5.999	5.964	5.912	5.858	5.803	5.746	5.717	5.688	5.658	5.628
5	6.608	5.786	5.409	5.192	5.050	4.950	4.876	4.818	4.772	4.735	4.678	4.619	4.558	4.496	4.464	4.431	4.398	4.365
6	5.987	5.143	4.757	4.534	4.387	4.284	4.207	4.147	4.099	4.060	4.000	3.938	3.874	3.808	3.774	3.740	3.705	3.669
7	5.591	4.737	4.347	4.120	3.972	3.866	3.787	3.726	3.677	3.637	3.575	3.511	3.445	3.376	3.340	3.304	3.267	3.230
8	5.318	4.459	4.066	3.838	3.688	3.581	3.500	3.438	3.388	3.347	3.284	3.218	3.150	3.079	3.043	3.005	2.967	2.928
9	5.117	4.256	3.863	3.633	3.482	3.374	3.293	3.230	3.179	3.137	3.073	3.006	2.936	2.864	2.826	2.787	2.748	2.707
10	4.965	4.103	3.708	3.478	3.326	3.217	3.135	3.072	3.020	2.978	2.913	2.845	2.774	2.700	2.661	2.621	2.580	2.538
11	4.844	3.982	3.587	3.357	3.204	3.095	3.012	2.948	2.896	2.854	2.788	2.719	2.646	2.570	2.531	2.490	2.448	2.404
12	4.747	3.885	3.490	3.259	3.106	2.996	2.913	2.849	2.796	2.753	2.687	2.617	2.544	2.466	2.426	2.384	2.341	2.296
13	4.667	3.806	3.411	3.179	3.025	2.915	2.832	2.767	2.714	2.671	2.604	2.533	2.459	2.380	2.339	2.297	2.252	2.206
14	4.600	3.739	3.344	3.112	2.958	2.848	2.764	2.699	2.646	2.602	2.534	2.463	2.388	2.308	2.266	2.223	2.178	2.131
15	4.543	3.682	3.287	3.056	2.901	2.790	2.707	2.641	2.588	2.544	2.475	2.403	2.328	2.247	2.204	2.160	2.114	2.066
16	4.494	3.634	3.239	3.007	2.852	2.741	2.657	2.591	2.538	2.494	2.425	2.352	2.276	2.194	2.151	2.106	2.059	2.010
17	4.451	3.592	3.197	2.965	2.810	2.699	2.614	2.548	2.494	2.450	2.381	2.308	2.230	2.148	2.104	2.058	2.011	1.960
18	4.414	3.555	3.160	2.928	2.773	2.661	2.577	2.510	2.456	2.412	2.342	2.269	2.191	2.107	2.063	2.017	1.968	1.917
19	4.381	3.522	3.127	2.895	2.740	2.628	2.544	2.477	2.423	2.378	2.308	2.234	2.155	2.071	2.026	1.980	1.930	1.878
20	4.351	3.493	3.098	2.866	2.711	2.599	2.514	2.447	2.393	2.348	2.278	2.203	2.124	2.039	1.994	1.946	1.896	1.843
21	4.325	3.467	3.072	2.840	2.685	2.573	2.488	2.420	2.366	2.321	2.250	2.176	2.096	2.010	1.965	1.916	1.866	1.812
22	4.301	3.443	3.049	2.817	2.661	2.549	2.464	2.397	2.342	2.297	2.226	2.151	2.071	1.984	1.938	1.889	1.838	1.783
23	4.279	3.422	3.028	2.796	2.640	2.528	2.442	2.375	2.320	2.275	2.204	2.128	2.048	1.961	1.914	1.865	1.813	1.757
24	4.260	3.403	3.009	2.776	2.621	2.508	2.423	2.355	2.300	2.255	2.183	2.108	2.027	1.939	1.892	1.842	1.790	1.733
30	4.171	3.316	2.922	2.690	2.534	2.421	2.334	2.266	2.211	2.165	2.092	2.015	1.932	1.841	1.792	1.740	1.683	1.622
40	4.085	3.232	2.839	2.606	2.449	2.336	2.249	2.180	2.124	2.077	2.003	1.924	1.839	1.744	1.693	1.637	1.577	1.509
60	4.001	3.150	2.758	2.525	2.368	2.254	2.167	2.097	2.040	1.993	1.917	1.836	1.748	1.649	1.594	1.534	1.467	1.389
120	3.920	3.072	2.680	2.447	2.290	2.175	2.087	2.016	1.959	1.910	1.834	1.750	1.659	1.554	1.495	1.429	1.352	1.254
∞	3.841	2.996	2.605	2.372	2.214	2.099	2.010	1.938	1.880	1.831	1.752	1.666	1.571	1.459	1.394	1.318	1.221	1.000

TABLE II Binomial Probabilities



Tabulated values are $\sum_{x=0}^{k} p(x)$. *(Computations are rounded at the third decimal place.)*

a. $n = 5$

k \ p	.01	.05	.10	.20	.30	.40	.50	.60	.70	.80	.90	.95	.99
0	.951	.774	.590	.328	.168	.078	.031	.010	.002	.000	.000	.000	.000
1	.999	.977	.919	.737	.528	.337	.188	.087	.031	.007	.000	.000	.000
2	1.000	.999	.991	.942	.837	.683	.500	.317	.163	.058	.009	.001	.000
3	1.000	1.000	1.000	.993	.969	.913	.812	.663	.472	.263	.081	.023	.001
4	1.000	1.000	1.000	1.000	.998	.990	.969	.922	.832	.672	.410	.226	.049

b. $n = 6$

k \ p	.01	.05	.10	.20	.30	.40	.50	.60	.70	.80	.90	.95	.99
0	.941	.735	.531	.262	.118	.047	.016	.004	.001	.000	.000	.000	.000
1	.999	.967	.886	.655	.420	.233	.109	.041	.011	.002	.000	.000	.000
2	1.000	.998	.984	.901	.744	.544	.344	.179	.070	.017	.001	.000	.000
3	1.000	1.000	.999	.983	.930	.821	.656	.456	.256	.099	.016	.002	.000
4	1.000	1.000	1.000	.998	.989	.959	.891	.767	.580	.345	.114	.033	.001
5	1.000	1.000	1.000	1.000	.999	.996	.984	.953	.882	.738	.469	.265	.059

c. $n = 7$

k \ p	.01	.05	.10	.20	.30	.40	.50	.60	.70	.80	.90	.95	.99
0	.932	.698	.478	.210	.082	.028	.008	.002	.000	.000	.000	.000	.000
1	.998	.956	.850	.577	.329	.159	.063	.019	.004	.000	.000	.000	.000
2	1.000	.996	.974	.852	.647	.420	.227	.096	.029	.005	.000	.000	.000
3	1.000	1.000	.997	.967	.874	.710	.500	.290	.126	.033	.003	.000	.000
4	1.000	1.000	1.000	.995	.971	.904	.773	.580	.353	.148	.026	.004	.000
5	1.000	1.000	1.000	1.000	.996	.981	.937	.841	.671	.423	.150	.044	.002
6	1.000	1.000	1.000	1.000	1.000	.998	.992	.972	.918	.790	.522	.302	.068

(continued)

TABLE II (*continued*)

d. $n = 8$

k \ p	.01	.05	.10	.20	.30	.40	.50	.60	.70	.80	.90	.95	.99
0	.923	.663	.430	.168	.058	.017	.004	.001	.000	.000	.000	.000	.000
1	.997	.943	.813	.503	.255	.106	.035	.009	.001	.000	.000	.000	.000
2	1.000	.994	.962	.797	.552	.315	.145	.050	.011	.001	.000	.000	.000
3	1.000	1.000	.995	.944	.806	.594	.363	.174	.058	.010	.000	.000	.000
4	1.000	1.000	1.000	.990	.942	.826	.637	.406	.194	.056	.005	.000	.000
5	1.000	1.000	1.000	.999	.989	.950	.855	.685	.448	.203	.038	.006	.000
6	1.000	1.000	1.000	1.000	.999	.991	.965	.894	.745	.497	.187	.057	.003
7	1.000	1.000	1.000	1.000	1.000	.999	.996	.983	.942	.832	.570	.337	.077

e. $n = 9$

k \ p	.01	.05	.10	.20	.30	.40	.50	.60	.70	.80	.90	.95	.99
0	.914	.630	.387	.134	.040	.010	.002	.000	.000	.000	.000	.000	.000
1	.997	.929	.775	.436	.196	.071	.020	.004	.000	.000	.000	.000	.000
2	1.000	.992	.947	.738	.463	.232	.090	.025	.004	.000	.000	.000	.000
3	1.000	.999	.992	.914	.730	.483	.254	.099	.025	.003	.000	.000	.000
4	1.000	1.000	.999	.980	.901	.733	.500	.267	.099	.020	.001	.000	.000
5	1.000	1.000	1.000	.997	.975	.901	.746	.517	.270	.086	.008	.001	.000
6	1.000	1.000	1.000	1.000	.996	.975	.910	.768	.537	.262	.053	.008	.000
7	1.000	1.000	1.000	1.000	1.000	.996	.980	.929	.804	.564	.225	.071	.003
8	1.000	1.000	1.000	1.000	1.000	1.000	.998	.990	.960	.866	.613	.370	.086

f. $n = 10$

k \ p	.01	.05	.10	.20	.30	.40	.50	.60	.70	.80	.90	.95	.99
0	.904	.599	.349	.107	.028	.006	.001	.000	.000	.000	.000	.000	.000
1	.996	.914	.736	.376	.149	.046	.011	.002	.000	.000	.000	.000	.000
2	1.000	.988	.930	.678	.383	.167	.055	.012	.002	.000	.000	.000	.000
3	1.000	.999	.987	.879	.650	.382	.172	.055	.011	.001	.000	.000	.000
4	1.000	1.000	.998	.967	.850	.633	.377	.166	.047	.006	.000	.000	.000
5	1.000	1.000	1.000	.994	.953	.834	.623	.367	.150	.033	.002	.000	.000
6	1.000	1.000	1.000	.999	.989	.945	.828	.618	.350	.121	.013	.001	.000
7	1.000	1.000	1.000	1.000	.998	.988	.945	.833	.617	.322	.070	.012	.000
8	1.000	1.000	1.000	1.000	1.000	.998	.989	.954	.851	.624	.264	.086	.004
9	1.000	1.000	1.000	1.000	1.000	1.000	.999	.994	.972	.893	.651	.401	.096

(*continued*)

TABLE II (*continued*)

g. $n = 15$

k \ p	.01	.05	.10	.20	.30	.40	.50	.60	.70	.80	.90	.95	.99
0	.860	.463	.206	.035	.005	.000	.000	.000	.000	.000	.000	.000	.000
1	.990	.829	.549	.167	.035	.005	.000	.000	.000	.000	.000	.000	.000
2	1.000	.964	.816	.398	.127	.027	.004	.000	.000	.000	.000	.000	.000
3	1.000	.995	.944	.648	.297	.091	.018	.002	.000	.000	.000	.000	.000
4	1.000	.999	.987	.838	.515	.217	.059	.009	.001	.000	.000	.000	.000
5	1.000	1.000	.998	.939	.722	.403	.151	.034	.004	.000	.000	.000	.000
6	1.000	1.000	1.000	.982	.869	.610	.304	.095	.015	.001	.000	.000	.000
7	1.000	1.000	1.000	.996	.950	.787	.500	.213	.050	.004	.000	.000	.000
8	1.000	1.000	1.000	.999	.985	.905	.696	.390	.131	.018	.000	.000	.000
9	1.000	1.000	1.000	1.000	.996	.966	.849	.597	.278	.061	.002	.000	.000
10	1.000	1.000	1.000	1.000	.999	.991	.941	.783	.485	.164	.013	.001	.000
11	1.000	1.000	1.000	1.000	1.000	.998	.982	.909	.703	.352	.056	.005	.000
12	1.000	1.000	1.000	1.000	1.000	1.000	.996	.973	.873	.602	.184	.036	.000
13	1.000	1.000	1.000	1.000	1.000	1.000	1.000	.995	.965	.833	.451	.171	.010
14	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	.995	.965	.794	.537	.140

h. $n = 20$

k \ p	.01	.05	.10	.20	.30	.40	.50	.60	.70	.80	.90	.95	.99
0	.818	.358	.122	.012	.001	.000	.000	.000	.000	.000	.000	.000	.000
1	.983	.736	.392	.069	.008	.001	.000	.000	.000	.000	.000	.000	.000
2	.999	.925	.677	.206	.035	.004	.000	.000	.000	.000	.000	.000	.000
3	1.000	.984	.867	.411	.107	.016	.001	.000	.000	.000	.000	.000	.000
4	1.000	.997	.957	.630	.238	.051	.006	.000	.000	.000	.000	.000	.000
5	1.000	1.000	.989	.804	.416	.126	.021	.002	.000	.000	.000	.000	.000
6	1.000	1.000	.998	.913	.608	.250	.058	.006	.000	.000	.000	.000	.000
7	1.000	1.000	1.000	.968	.772	.416	.132	.021	.001	.000	.000	.000	.000
8	1.000	1.000	1.000	.990	.887	.596	.252	.057	.005	.000	.000	.000	.000
9	1.000	1.000	1.000	.997	.952	.755	.412	.128	.017	.001	.000	.000	.000
10	1.000	1.000	1.000	.999	.983	.872	.588	.245	.048	.003	.000	.000	.000
11	1.000	1.000	1.000	1.000	.995	.943	.748	.404	.113	.010	.000	.000	.000
12	1.000	1.000	1.000	1.000	.999	.979	.868	.584	.228	.032	.000	.000	.000
13	1.000	1.000	1.000	1.000	1.000	.994	.942	.750	.392	.087	.002	.000	.000
14	1.000	1.000	1.000	1.000	1.000	.998	.979	.874	.584	.196	.011	.000	.000
15	1.000	1.000	1.000	1.000	1.000	1.000	.994	.949	.762	.370	.043	.003	.000
16	1.000	1.000	1.000	1.000	1.000	1.000	.999	.984	.893	.589	.133	.016	.000
17	1.000	1.000	1.000	1.000	1.000	1.000	1.000	.996	.965	.794	.323	.075	.001
18	1.000	1.000	1.000	1.000	1.000	1.000	1.000	.999	.992	.931	.608	.264	.017
19	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	.999	.988	.878	.642	.182

(*continued*)

TABLE XIV Critical Values of T_L and T_U for the Wilcoxon Rank Sum Test: Independent Samples

Test statistic is the rank sum associated with the smaller sample (if equal sample sizes, either rank sum can be used).

a. $\alpha = .025$ one-tailed; $\alpha = .05$ two-tailed

n_2 \ n_1	3		4		5		6		7		8		9		10	
	T_L	T_U	T_L	T_U	T_L	T_U	T_L	T_U	T_L	T_U	T_L	T_U	T_L	T_U	T_L	T_U
3	5	16	6	18	6	21	7	23	7	26	8	28	8	31	9	33
4	6	18	11	25	12	28	12	32	13	35	14	38	15	41	16	44
5	6	21	12	28	18	37	19	41	20	45	21	49	22	53	24	56
6	7	23	12	32	19	41	26	52	28	56	29	61	31	65	32	70
7	7	26	13	35	20	45	28	56	37	68	39	73	41	78	43	83
8	8	28	14	38	21	49	29	61	39	73	49	87	51	93	54	98
9	8	31	15	41	22	53	31	65	41	78	51	93	63	108	66	114
10	9	33	16	44	24	56	32	70	43	83	54	98	66	114	79	131

b. $\alpha = .05$ one-tailed; $\alpha = .10$ two-tailed

n_2 \ n_1	3		4		5		6		7		8		9		10	
	T_L	T_U	T_L	T_U	T_L	T_U	T_L	T_U	T_L	T_U	T_L	T_U	T_L	T_U	T_L	T_U
3	6	15	7	17	7	20	8	22	9	24	9	27	10	29	11	31
4	7	17	12	24	13	27	14	30	15	33	16	36	17	39	18	42
5	7	20	13	27	19	36	20	40	22	43	24	46	25	50	26	54
6	8	22	14	30	20	40	28	50	30	54	32	58	33	63	35	67
7	9	24	15	33	22	43	30	54	39	66	41	71	43	76	46	80
8	9	27	16	36	24	46	32	58	41	71	52	84	54	90	57	95
9	10	29	17	39	25	50	33	63	43	76	54	90	66	105	69	111
10	11	31	18	42	26	54	35	67	46	80	57	95	69	111	83	127

Source: From F. Wilcoxon, and R. A. Wilcox. "Some Rapid Approximate Statistical Procedures," 1964, 20–23. Courtesy of Lederle Laboratories Division of American Cyanamid Company, Madison, NJ.